

FOUNDATION CALCULATION SHEET

One-Stop Solution for Foundation



TITLE		DESCRIPTION				
PROJECT/JOB NO.		12-1812 (121-D)				
PROJECT/JOB NAME		PUSRI IIB				
CLIENT NAME		PT PUSRI PLB				
SITE NAME		PALEMBANG, SOUTH SUMATERA INDONESIA				
DOCUMENT NO.						
REFERENCE NO.						
STRUCTURE NAME		121-D Rev 2_by FST_SF=3_1				
LOAD COMBINATION GROUP						

REV	DATE	DESCRIPTION	PREP'D	CHK'D	APPR'D	APPR'D

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1. GENERAL

1.1 CODE & STANDARD

Items	Description
Design Code	American Concrete Institute (ACI 318) [Metric]
Horizontal Force for Wind	NONE
Horizontal Force for Seismic	NONE
Unit System	Input : MKS, Output : MKS, Calculation Unit : IMPERIAL

1.2 MATERIALS & UNIT WEIGHT

Items	Value
Concrete (f'_c : compressive strength)	210.000 kgf/cm ²
Lean Concrete (Lf'_c : compressive strength)	130.000 kgf/cm ²
Reinforcement (D10 ~ D16 , yield strength)	40000.000 kgf/cm ²
Reinforcement (D19 ~ , yield strength)	40000.000 kgf/cm ²
Rs (Soil unit weight)	2.200 ton/m ³
Rc (Concrete unit weight)	2.400 ton/m ³
Es (Steel Modulus of Elasticity)	2.039×10^6 kgf/cm ²
Ec (Concrete Modulus of Elasticity)	217500.000 kgf/cm ²

- Pile Capacity

Items	Value
Pile Name	D600_L=18 m
Footing List	F6
Diameter	600 mm
Length	18 m
Thick	12 mm
Shape	Circle
Capacity (H_a , U_a , V_a)	23.8 , 26 , 225 tonf

Items	Value
Pile Name	D600_L=24 m
Footing List	F6
Diameter	600 mm
Length	24 m
Thick	12 mm
Shape	Circle
Capacity (H_a , U_a , V_a)	23.8 , 54 , 135 tonf

Items	Value
Pile Name	D600_L=26/27m
Footing List	F6
Diameter	600 mm
Length	26 m
Thick	12 mm
Shape	Circle



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Capacity (Ha , Ua , Va)	23.8 , 64.8 , 166.5 tonf
Items	Value
Pile Name	D600_L=30 M
Footing List	F6
Diameter	600 mm
Length	30 m
Thick	12 mm
Shape	Circle
Capacity (Ha , Ua , Va)	23.8 , 91.3 , 225 tonf
Items	Value
Pile Name	D600_L=30m_per
Footing List	F6
Diameter	600 mm
Length	30 m
Thick	12 mm
Shape	Circle
Capacity (Ha , Ua , Va)	23.8 , 96 , 225 tonf
Items	Value
Pile Name	D600_L=26/27m_per
Footing List	F6
Diameter	600 mm
Length	27 m
Thick	12 mm
Shape	Circle
Capacity (Ha , Ua , Va)	23.8 , 69 , 166.5 tonf
Items	Value
Pile Name	D600_L30m_per_tension
Footing List	F6
Diameter	600 mm
Length	30 m
Thick	12 mm
Shape	Circle
Capacity (Ha , Ua , Va)	23.8 , 96.3 , 225 tonf

1.3 SUBSOIL CONDITION & SAFETY FACTORS

Items	Description
Allowable Increase of Soil (Wind)	33 %
Allowable Increase of Soil (Seismic)	33 %
Allowable Increase of Soil (Test)	20 %
Allowable Increase of Pile Horizontal (Wind)	33 %
Allowable Increase of Pile Horizontal (Seismic)	33 %
Allowable Increase of Pile Horizontal (Test)	20 %
Allowable Increase of Pile Vertical (Wind)	33 %
Allowable Increase of Pile Vertical (Seismic)	33 %



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Allowable Increase of Pile Vertical (Test)	20 %
Allowable Increase of Pile Uplift (Wind)	33 %
Allowable Increase of Pile Uplift (Seismic)	33 %
Allowable Increase of Pile Uplift (Test)	20 %
Safety factor against overturning for OVM1(FO1)	1.5
Safety factor against overturning for OVM2(FO2)	1.5
Safety factor against overturning for OVM3(FO3)	1.5
Safety factor against overturning for OVM4(FO4)	1.5
Safety factor against sliding for the SL1(FS1)	1.5
Safety factor against sliding for the SL2(FS2)	1.5
Safety factor against sliding for the SL3(FS3)	1.5
Safety factor against sliding for the SL4(FS4)	1.5
Friction factor (μ)	0.35

1.4 LOAD COMBINATION

Index	Load Case Name	Load Case Description
1	SW	SELF WEIGHT
2	DL	DEAD LOAD
3	E(e) : Equipment Load (	E(e) : Equipment Load (Erection)
4	E(o) : Equipment Load (	E(o) : Equipment Load (Operation)
5	E(t) : Equipment Load (	E(t) : Equipment Load (Test)
6	P(e) : Piping Load	P(e) : Piping Load (Erection)
7	P(o) : Piping Load	P(o) : Piping Load (Operation)
8	P(t) : Piping Load (Test)	P(t) : Piping Load (Test)
9	W(x) : Severe Load (	W(x) : Severe Load (Wind X)
10	-w(x) : Severe Load (	-w(x) : Severe Load (Wind -x)
11	W(z) : Severe Load (	W(z) : Severe Load (Wind Z)
12	-w(z) : Severe Load (	-w(z) : Severe Load (Wind -z)
13	V(x) : Severe Load (	V(x) : Severe Load (Seismic X)
14	V(z) : Severe Load (	V(z) : Severe Load (Seismic Z)
15	V(x) : Severe Load (	V(x) : Severe Load (Seismic X) Erection
16	V(z) : Severe Load (	V(z) : Severe Load (Seismic Z) Erection

Comb . ID	Load Combination for stability
101	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection)
102	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + .6 W(x) : Severe Load (Wind X)
103	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + .6 -w(x) : Severe Load (Wind -x)
104	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + .6 W(z) : Severe Load (Wind Z)
105	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + .6 -w(z) : Severe Load (Wind -z)
106	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + .7 V(x) : Severe Load (Seismic X) Erection + .21 V(z) : Severe Load (Seismic Z) Erection
107	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + .7 V(x) : Severe Load (Seismic X) Erection + -.21 V(z) : Severe Load (Seismic Z) Erection



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108	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + -.7 V(x) : Severe Load (Seismic X) Erection + .21 V(z) : Severe Load (Seismic Z) Erection
109	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + -.7 V(x) : Severe Load (Seismic X) Erection + .21 V(z) : Severe Load (Seismic Z) Erection
110	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + .21 V(x) : Severe Load (Seismic X) Erection + .7 V(z) : Severe Load (Seismic Z) Erection
111	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + .21 V(x) : Severe Load (Seismic X) Erection + -.7 V(z) : Severe Load (Seismic Z) Erection
112	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + -.21 V(x) : Severe Load (Seismic X) Erection + .7 V(z) : Severe Load (Seismic Z) Erection
113	1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection) + -.21 V(x) : Severe Load (Seismic X) Erection + -.7 V(z) : Severe Load (Seismic Z) Erection
114	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + .6 W(x) : Severe Load (Wind X)
115	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + .6 -w(x) : Severe Load (Wind -x)
116	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + .6 W(z) : Severe Load (Wind Z)
117	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + .6 -w(z) : Severe Load (Wind -z)
118	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + .7 V(x) : Severe Load (Seismic X) Erection + .21 V(z) : Severe Load (Seismic Z) Erection
119	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + .7 V(x) : Severe Load (Seismic X) Erection + -.21 V(z) : Severe Load (Seismic Z) Erection
120	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + -.7 V(x) : Severe Load (Seismic X) Erection + .21 V(z) : Severe Load (Seismic Z) Erection
121	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + -.7 V(x) : Severe Load (Seismic X) Erection + -.21 V(z) : Severe Load (Seismic Z) Erection
122	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + .21 V(x) : Severe Load (Seismic X) Erection + .7 V(z) : Severe Load (Seismic Z) Erection
123	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + .21 V(x) : Severe Load (Seismic X) Erection + -.7 V(z) : Severe Load (Seismic Z) Erection
124	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + -.21 V(x) : Severe Load (Seismic X) Erection + .7 V(z) : Severe Load (Seismic Z) Erection
125	.6 SW + .6 DL + .6 E(e) : Equipment Load (Erection) + .6 P(e) : Piping Load (Erection) + -.21 V(x) : Severe Load (Seismic X) Erection + -.7 V(z) : Severe Load (Seismic Z) Erection
126	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation)
127	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .45 W(x) : Severe Load (Wind X)
128	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .45 -w(x) : Severe Load (Wind -x)
129	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .45 W(z) : Severe Load (Wind Z)
130	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .45 -w(z) : Severe Load (Wind -z)
131	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .525 V(x) : Severe Load (Seismic X) + .158 V(z) : Severe Load (Seismic Z)
132	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .525 V(x) : Severe Load (Seismic X) + -.158 V(z) : Severe Load (Seismic Z)
133	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.525 V(x) : Severe Load (Seismic X) + .158 V(z) : Severe Load (Seismic Z)
134	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.525 V(x) : Severe Load (Seismic X) + -.158 V(z) : Severe Load (Seismic Z)



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135	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .158 V(x) : Severe Load (Seismic X) + .525 V(z) : Severe Load (Seismic Z)
136	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .158 V(x) : Severe Load (Seismic X) + -.525 V(z) : Severe Load (Seismic Z)
137	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.158 V(x) : Severe Load (Seismic X) + .525 V(z) : Severe Load (Seismic Z)
138	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.158 V(x) : Severe Load (Seismic X) + -.525 V(z) : Severe Load (Seismic Z)
139	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + .6 W(x) : Severe Load (Wind X)
140	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + .6 -w(x) : Severe Load (Wind -x)
141	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + .6 W(z) : Severe Load (Wind Z)
142	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + .6 -w(z) : Severe Load (Wind -z)
143	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + .7 V(x) : Severe Load (Seismic X) + .21 V(z) : Severe Load (Seismic Z)
144	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + .7 V(x) : Severe Load (Seismic X) + -.21 V(z) : Severe Load (Seismic Z)
145	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + -.7 V(x) : Severe Load (Seismic X) + .21 V(z) : Severe Load (Seismic Z)
146	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + -.7 V(x) : Severe Load (Seismic X) + -.21 V(z) : Severe Load (Seismic Z)
147	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + .21 V(x) : Severe Load (Seismic X) + .7 V(z) : Severe Load (Seismic Z)
148	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + .21 V(x) : Severe Load (Seismic X) + -.7 V(z) : Severe Load (Seismic Z)
149	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + -.21 V(x) : Severe Load (Seismic X) + .7 V(z) : Severe Load (Seismic Z)
150	.6 SW + .6 DL + .6 E(o) : Equipment Load (Operation) + .6 P(o) : Piping Load (Operation) + -.21 V(x) : Severe Load (Seismic X) + -.7 V(z) : Severe Load (Seismic Z)
151	1.0 SW + 1.0 DL + 1.0 E(t) : Equipment Load (Test) + 1.0 P(t) : Piping Load (Test)
152	1.0 SW + 1.0 DL + 1.0 E(t) : Equipment Load (Test) + 1.0 P(t) : Piping Load (Test) + .45 W(x) : Severe Load (Wind X)
153	1.0 SW + 1.0 DL + 1.0 E(t) : Equipment Load (Test) + 1.0 P(t) : Piping Load (Test) + .45 -w(x) : Severe Load (Wind -x)
154	1.0 SW + 1.0 DL + 1.0 E(t) : Equipment Load (Test) + 1.0 P(t) : Piping Load (Test) + .45 W(z) : Severe Load (Wind Z)
155	1.0 SW + 1.0 DL + 1.0 E(t) : Equipment Load (Test) + 1.0 P(t) : Piping Load (Test) + .45 -w(z) : Severe Load (Wind -z)
156	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .6 W(x) : Severe Load (Wind X)
157	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .6 -w(x) : Severe Load (Wind -x)
158	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .6 W(z) : Severe Load (Wind Z)
159	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .6 -w(z) : Severe Load (Wind -z)
160	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .7 V(x) : Severe Load (Seismic X) + .21 V(z) : Severe Load (Seismic Z)
161	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .7 V(x) : Severe Load (Seismic X) + -.21 V(z) : Severe Load (Seismic Z)



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162	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.7 V(x) : Severe Load (Seismic X) + .21 V(z) : Severe Load (Seismic Z)
163	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.7 V(x) : Severe Load (Seismic X) + -.21 V(z) : Severe Load (Seismic Z)
164	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .21 V(x) : Severe Load (Seismic X) + .7 V(z) : Severe Load (Seismic Z)
165	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .21 V(x) : Severe Load (Seismic X) + -.7 V(z) : Severe Load (Seismic Z)
166	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.21 V(x) : Severe Load (Seismic X) + .7 V(z) : Severe Load (Seismic Z)
167	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.21 V(x) : Severe Load (Seismic X) + -.7 V(z) : Severe Load (Seismic Z)
401	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .6 W(x) : Severe Load (Wind X)
402	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .6 -w(x) : Severe Load (Wind -x)
403	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .6 W(z) : Severe Load (Wind Z)
404	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .6 -w(z) : Severe Load (Wind -z)
405	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .7 V(x) : Severe Load (Seismic X) + .21 V(z) : Severe Load (Seismic Z)
406	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .7 V(x) : Severe Load (Seismic X) + -.21 V(z) : Severe Load (Seismic Z)
407	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.7 V(x) : Severe Load (Seismic X) + .21 V(z) : Severe Load (Seismic Z)
408	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.7 V(x) : Severe Load (Seismic X) + -.21 V(z) : Severe Load (Seismic Z)
409	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .21 V(x) : Severe Load (Seismic X) + .7 V(z) : Severe Load (Seismic Z)
410	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + .21 V(x) : Severe Load (Seismic X) + -.7 V(z) : Severe Load (Seismic Z)
411	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.21 V(x) : Severe Load (Seismic X) + .7 V(z) : Severe Load (Seismic Z)
412	1.0 SW + 1.0 DL + 1.0 E(o) : Equipment Load (Operation) + 1.0 P(o) : Piping Load (Operation) + -.21 V(x) : Severe Load (Seismic X) + -.7 V(z) : Severe Load (Seismic Z)
Comb . ID	Load Combination for Reinforcement
201	1.4 SW + 1.4 DL + 1.4 E(e) : Equipment Load (Erection) + 1.4 P(e) : Piping Load (Erection)
202	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + 1.0 W(x) : Severe Load (Wind X)
203	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + 1.0 -w(x) : Severe Load (Wind -x)
204	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + 1.0 W(z) : Severe Load (Wind Z)
205	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + 1.0 -w(z) : Severe Load (Wind -z)
206	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + 1.0 V(x) : Severe Load (Seismic X) + .3 V(z) : Severe Load (Seismic Z)
207	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + 1.0 V(x) : Severe Load (Seismic X) Erection + -.3 V(z) : Severe Load (Seismic Z) Erection
208	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + -1.0 V(x) : Severe Load (Seismic X) Erection + .3 V(z) : Severe Load (Seismic Z) Erection



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209	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + -1.0 V(x) : Severe Load (Seismic X) Erection + -.3 V(z) : Severe Load (Seismic Z) Erection
210	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + .3 V(x) : Severe Load (Seismic X) Erection + 1.0 V(z) : Severe Load (Seismic Z) Erection
211	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + .3 V(x) : Severe Load (Seismic X) Erection + -1.0 V(z) : Severe Load (Seismic Z) Erection
212	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + -.3 V(x) : Severe Load (Seismic X) Erection + 1.0 V(z) : Severe Load (Seismic Z) Erection
213	1.2 SW + 1.2 DL + 1.2 E(e) : Equipment Load (Erection) + 1.2 P(e) : Piping Load (Erection) + -.3 V(x) : Severe Load (Seismic X) Erection + -1.0 V(z) : Severe Load (Seismic Z) Erection
226	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + 1.0 W(x) : Severe Load (Wind X)
227	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + 1.0 -w(x) : Severe Load (Wind -x)
228	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + 1.0 W(z) : Severe Load (Wind Z)
229	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + 1.0 -w(z) : Severe Load (Wind -z)
230	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + 1.0 V(x) : Severe Load (Seismic X) Erection + .3 V(z) : Severe Load (Seismic Z) Erection
231	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + 1.0 V(x) : Severe Load (Seismic X) Erection + -.3 V(z) : Severe Load (Seismic Z) Erection
232	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + -1.0 V(x) : Severe Load (Seismic X) Erection + .3 V(z) : Severe Load (Seismic Z) Erection
233	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + -1.0 V(x) : Severe Load (Seismic X) Erection + -.3 V(z) : Severe Load (Seismic Z) Erection
234	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + .3 V(x) : Severe Load (Seismic X) Erection + 1.0 V(z) : Severe Load (Seismic Z) Erection
235	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + .3 V(x) : Severe Load (Seismic X) Erection + -1.0 V(z) : Severe Load (Seismic Z) Erection
236	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + -.3 V(x) : Severe Load (Seismic X) Erection + 1.0 V(z) : Severe Load (Seismic Z) Erection
237	.9 SW + .9 DL + .9 E(e) : Equipment Load (Erection) + .9 P(e) : Piping Load (Erection) + -.3 V(x) : Severe Load (Seismic X) Erection + -1.0 V(z) : Severe Load (Seismic Z) Erection
238	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation)
239	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + 1.0 W(x) : Severe Load (Wind X)
240	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + 1.0 -w(x) : Severe Load (Wind -x)
241	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + 1.0 W(z) : Severe Load (Wind Z)
242	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + 1.0 -w(z) : Severe Load (Wind -z)
243	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + 1.0 V(x) : Severe Load (Seismic X) + .3 V(z) : Severe Load (Seismic Z)
244	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + 1.0 V(x) : Severe Load (Seismic X) + -.3 V(z) : Severe Load (Seismic Z)
245	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + -1.0 V(x) : Severe Load (Seismic X) + .3 V(z) : Severe Load (Seismic Z)
246	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + -1.0 V(x) : Severe Load (Seismic X) + -.3 V(z) : Severe Load (Seismic Z)
247	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + .3 V(x) : Severe Load (Seismic X) + 1.0 V(z) : Severe Load (Seismic Z)



Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

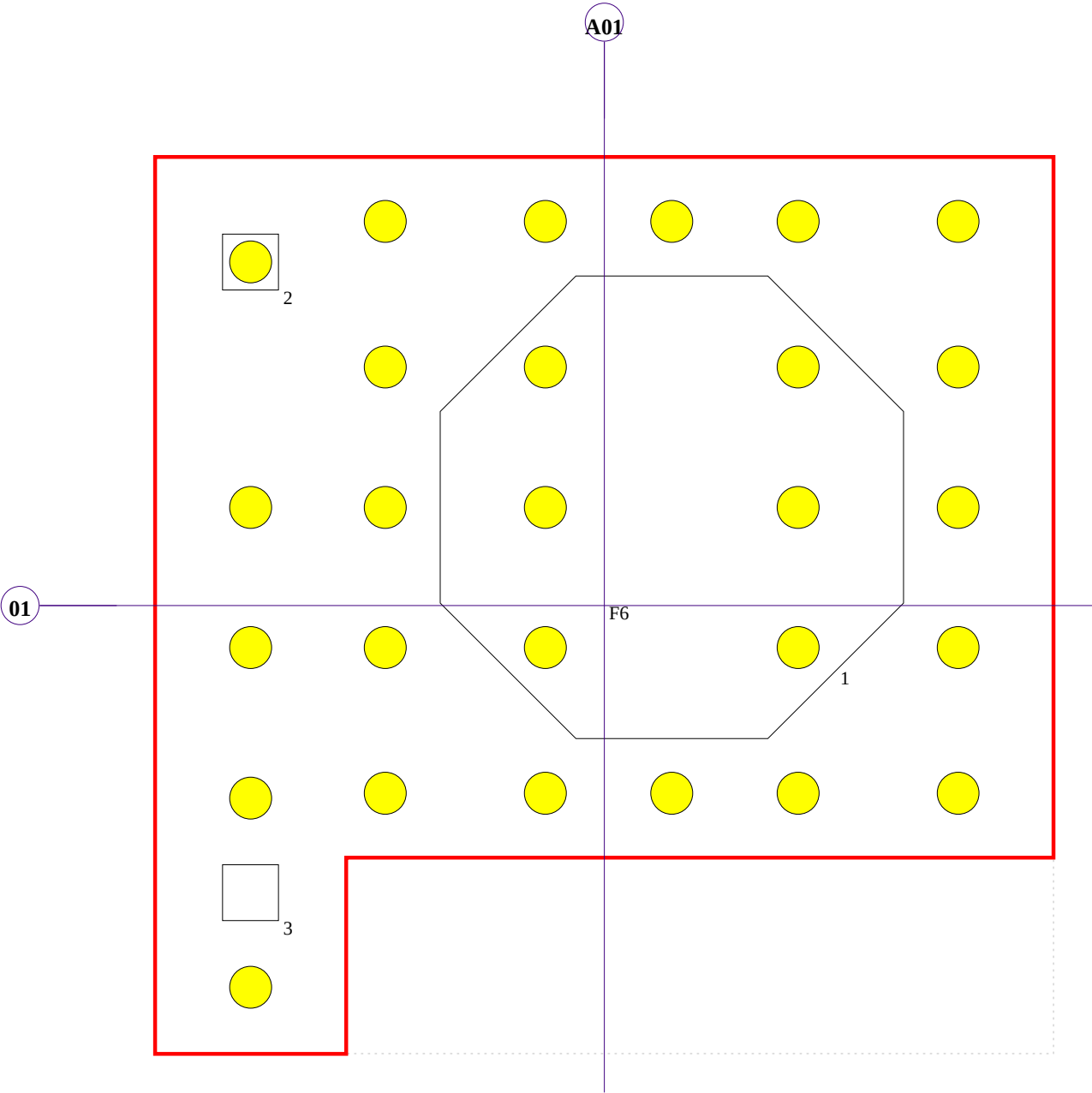
Project No. : 12-1812

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248	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + .3 V(x) : Severe Load (Seismic X) + -1.0 V(z) : Severe Load (Seismic Z)
249	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + -.3 V(x) : Severe Load (Seismic X) + 1.0 V(z) : Severe Load (Seismic Z)
250	1.2 SW + 1.2 DL + 1.2 E(o) : Equipment Load (Operation) + 1.2 P(o) : Piping Load (Operation) + -.3 V(x) : Severe Load (Seismic X) + -1.0 V(z) : Severe Load (Seismic Z)
256	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + 1.0 W(x) : Severe Load (Wind X)
257	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + 1.0 -w(x) : Severe Load (Wind -x)
258	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + 1.0 W(z) : Severe Load (Wind Z)
259	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + 1.0 -w(z) : Severe Load (Wind -z)
260	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + 1.0 V(x) : Severe Load (Seismic X) + .3 V(z) : Severe Load (Seismic Z)
261	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + 1.0 V(x) : Severe Load (Seismic X) + -.3 V(z) : Severe Load (Seismic Z)
262	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + -1.0 V(x) : Severe Load (Seismic X) + .3 V(z) : Severe Load (Seismic Z)
263	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + -1.0 V(x) : Severe Load (Seismic X) + -.3 V(z) : Severe Load (Seismic Z)
264	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + .3 V(x) : Severe Load (Seismic X) + 1.0 V(z) : Severe Load (Seismic Z)
265	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + .3 V(x) : Severe Load (Seismic X) + -1.0 V(z) : Severe Load (Seismic Z)
266	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + -.3 V(x) : Severe Load (Seismic X) + 1.0 V(z) : Severe Load (Seismic Z)
267	.9 SW + .9 DL + .9 E(o) : Equipment Load (Operation) + .9 P(o) : Piping Load (Operation) + -.3 V(x) : Severe Load (Seismic X) + -1.0 V(z) : Severe Load (Seismic Z)
268	1.2 SW + 1.2 DL + 1.2 E(t) : Equipment Load (Test) + 1.2 P(t) : Piping Load (Test)
269	1.2 SW + 1.2 DL + 1.2 E(t) : Equipment Load (Test) + 1.2 P(t) : Piping Load (Test) + 1.2 W(x) : Severe Load (Wind X)
270	1.2 SW + 1.2 DL + 1.2 E(t) : Equipment Load (Test) + 1.2 P(t) : Piping Load (Test) + 1.2 -w(x) : Severe Load (Wind -x)
271	1.2 SW + 1.2 DL + 1.2 E(t) : Equipment Load (Test) + 1.2 P(t) : Piping Load (Test) + 1.2 W(z) : Severe Load (Wind Z)
272	1.2 SW + 1.2 DL + 1.2 E(t) : Equipment Load (Test) + 1.2 P(t) : Piping Load (Test) + 1.2 -w(z) : Severe Load (Wind -z)

2. DRAWING



REFERENCE DWGS

NO.	DWG NO.	DWG TITLE

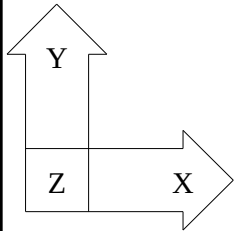
NOTES

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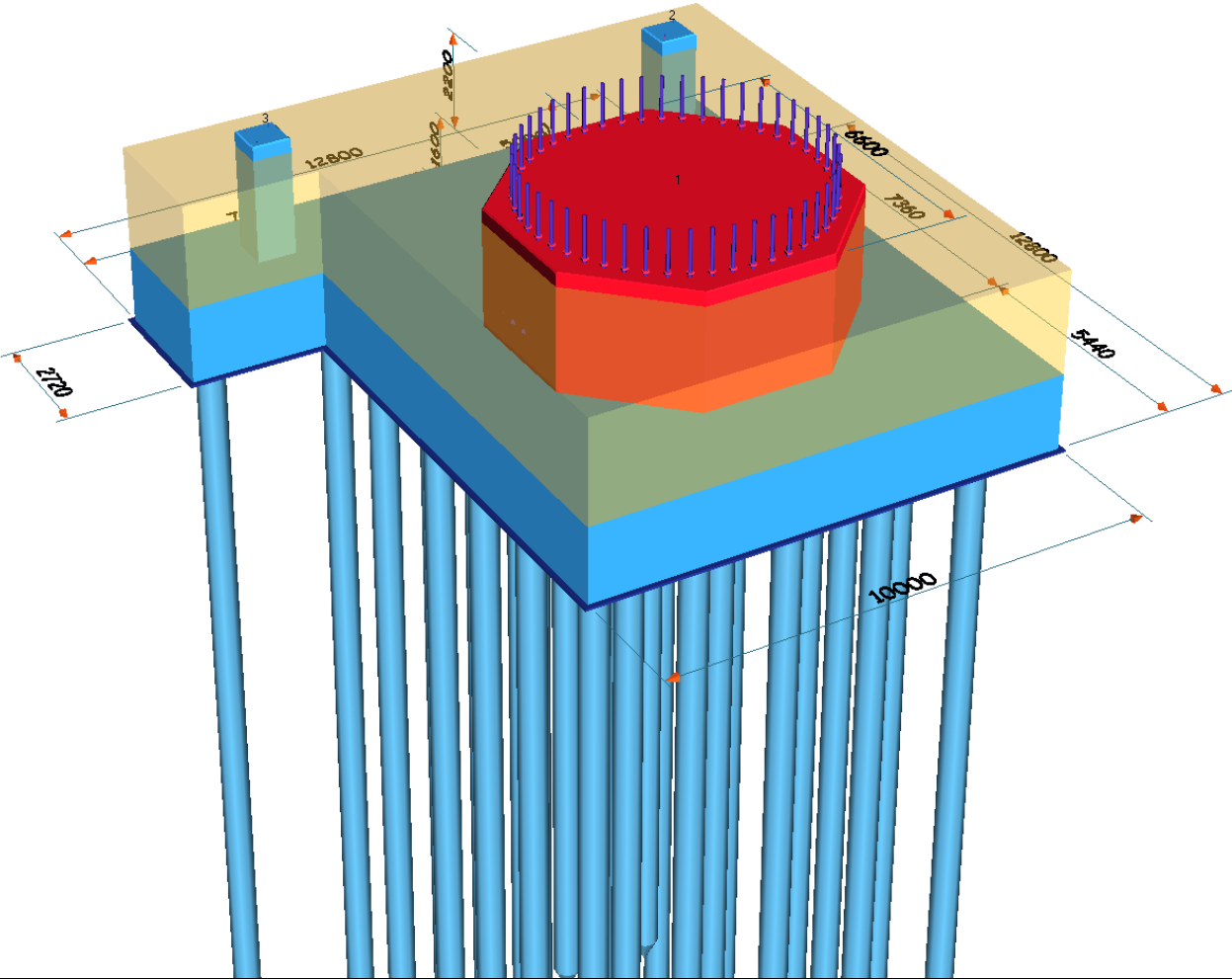
PUSRI IIB PROJECT

FOUNDATION LOCATION PLAN
121-D Rev 2_by FST_SF=3_1

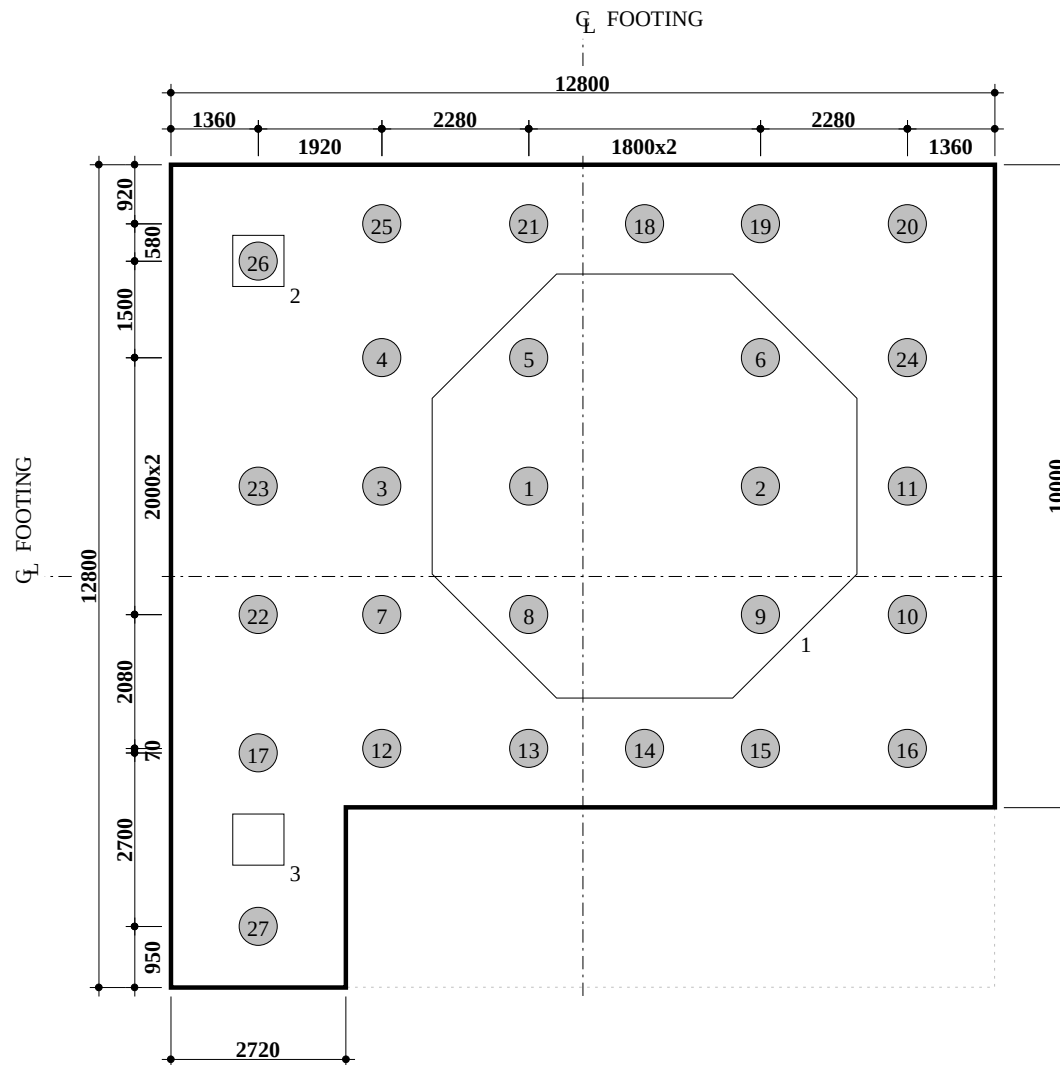
SQUAD CHECK						
	PROCESS	PIPING	VESSELS	STRUCT.	ELEC.	INST.
SCALE		JOB NO.		MICROFILM NO.		
AS SHOWN		12-1812 (121-D)				



2.2 DETAIL SKETCH



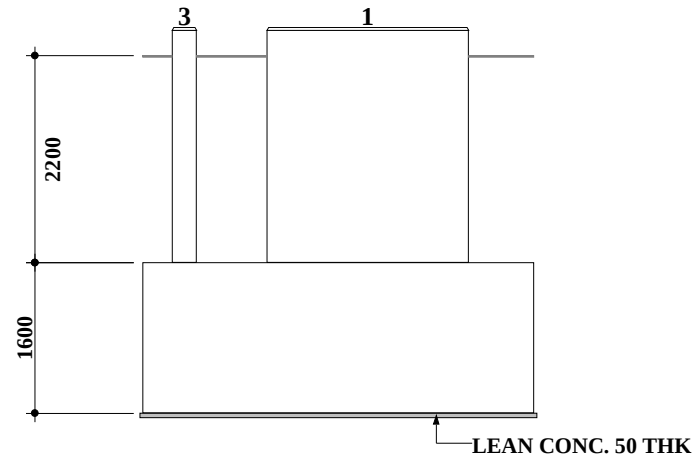
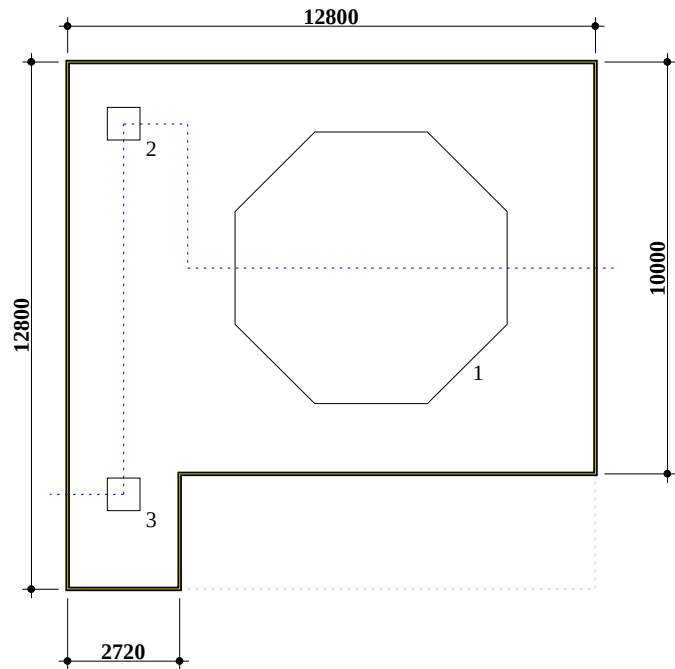
OUTPUT UNIT : mm



REFERENCE DWGS							
NO.	DWG NO.	DWG TITLE					
N O T E S							
* PILE 6-¥ö600 D600_L=18 m 1-¥ö600 D600_L=24 m 1-¥ö600 D600_L=26/27m 1-¥ö600 D600_L=30 M 14-¥ö600 D600_L=30m_per 3-¥ö600 D600_L=26/27m_per 1-¥ö600 D600_L30m_per_tension * ANCHOR BOLT 1X48-M100 ANC. BOLTS (TYPE TYPE LS) * OUTPUT UNIT : mm							
PUSRI IIB PROJECT							
FOUNDATION DETAIL FOR F6							
SQUAD CHECK							
	PROCESS	PIPING	VESSELS	STRUCT.	ELEC.	INST.	
PD							
SCALE AS SHOWN		JOB NO. 12-1812 (121-D)		MICROFILM NO.			



REFERENCE DWGS							
NO.	DWG NO.			DWG TITLE			
NOTES							
* PILE 6-Y5600 D600_L=18 m 1-Y5600 D600_L=24 m 1-Y5600 D600_L=26/27m 1-Y5600 D600_L=30 M 14-Y5600 D600_L=30m_per 3-Y5600 D600_L=26/27m_per 1-Y5600 D600_L30m_per_tension * ANCHOR BOLT 1X48-M100 ANC. BOLTS (TYPE TYPE LS) * OUTPUT UNIT : mm							
PUSRI IIB PROJECT							
FOUNDATION DETAIL FOR F6							
SQUAD CHECK							
	PROCESS	PIPING	VESSELS	STRUCT.	ELEC.	INST.	
	SCALE		JOB NO.		MICROFILM NO.		
PD	AS SHOWN		12-1812 (121-D)				



ELEVATION (S1) - X

Item No.			
1		2	3



Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

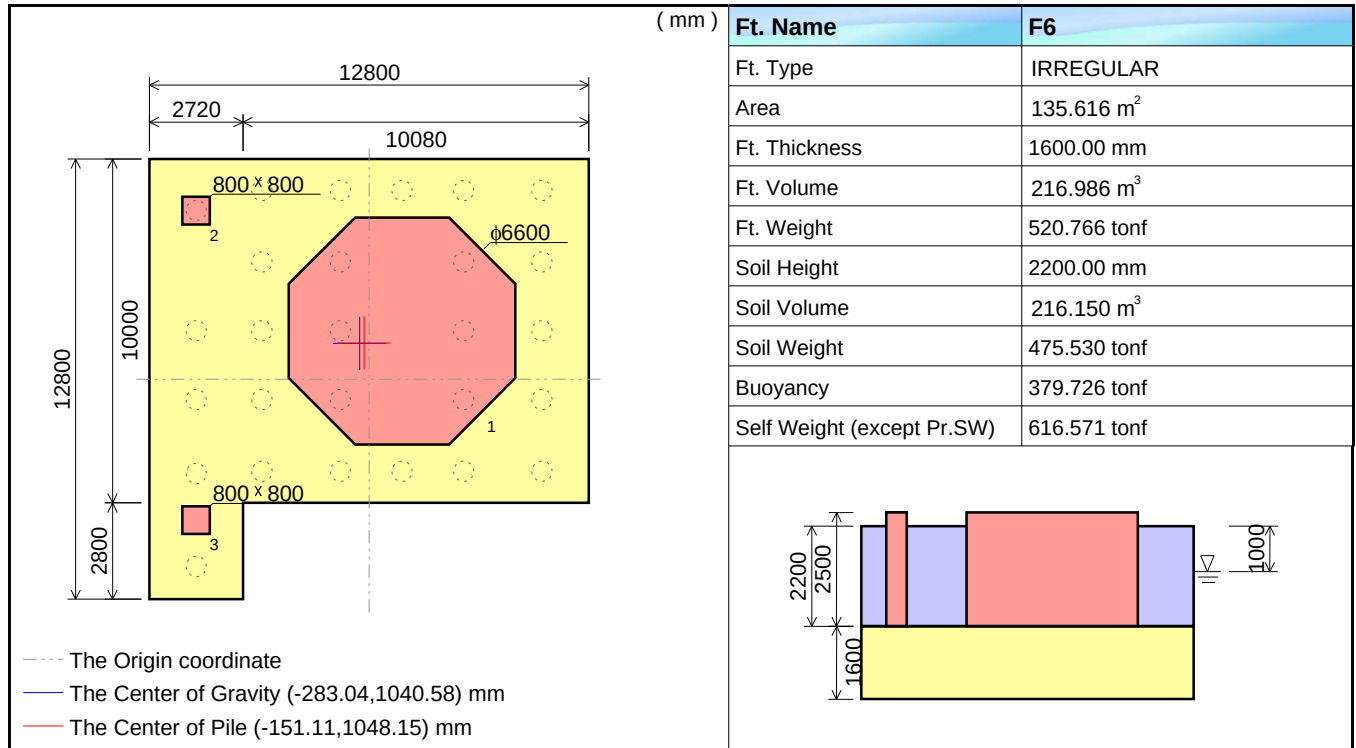
Project No. : 12-1812

Client : PT PUSRI PLB

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3. FOUNDATION DATA

3.1 FOOTING AND SECTION DATA



► Section Data

Ft.Name	Direction	Ft. Volume	Soil Volume	Pier Wt
F6	All Direct	216.986 m ³	216.150 m ³	224.198 tonf
Sec.Name	Section Area	Ft. Weight	Soil Weight	Total Weight
S1	135.616 m ²	520.766 tonf	475.530 tonf	1220.494 tonf

3.2 PIER DATA

Off X , Off Y is offset position from the Center of the footing

If Pier Shape is Circle or Circle wall, PI is a Diameter. and Pw is a Inner Diameter

Area is pier concrete area

Weight is pier and inner soil weight in case circle wall except Tank1 Type(Circle Ring Footing Shape)

Unit(Length : mm , Weight : tonf , Area : m²)

Ft.Name	Pr.Name	Shape	PI	Pw	Ph	Area	Weight	Off X	Off Y
F6	1	Octagonal	6600.000	6600.000	2500.000	36.086	216.518	960.000	1400.000
	2	Rectangle	800.000	800.000	2500.000	0.640	3.840	-5040.000	4900.000
	3	Rectangle	800.000	800.000	2500.000	0.640	3.840	-5040.000	-4100.000

3.3 LOAD CASE

	Input the point loads in the global coordinate system direction. Positive directions of moments (shown in the sketch) are based on the right hand rule.
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Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

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Index	Load Case Name	Load Case Description
1	SW	SELF WEIGHT
2	DL	DEAD LOAD
3	E(e) : Equipment Load (	E(e) : Equipment Load (Erection)
4	E(o) : Equipment Load (	E(o) : Equipment Load (Operation)
5	E(t) : Equipment Load (	E(t) : Equipment Load (Test)
6	P(e) : Piping Load	P(e) : Piping Load (Erection)
7	P(o) : Piping Load	P(o) : Piping Load (Operation)
8	P(t) : Piping Load (Test)	P(t) : Piping Load (Test)
9	W(x) : Severe Load (	W(x) : Severe Load (Wind X)
10	-w(x) : Severe Load (	-w(x) : Severe Load (Wind -x)
11	W(z) : Severe Load (	W(z) : Severe Load (Wind Z)
12	-w(z) : Severe Load (	-w(z) : Severe Load (Wind -z)
13	V(x) : Severe Load (	V(x) : Severe Load (Seismic X)
14	V(z) : Severe Load (	V(z) : Severe Load (Seismic Z)
15	V(x) : Severe Load (	V(x) : Severe Load (Seismic X) Erection
16	V(z) : Severe Load (	V(z) : Severe Load (Seismic Z) Erection

Unit(tonf , tonf-m)

Ft.Name	Pr.Name	Load Case	Fx	Fy	Fz	Mx	My
F6	1	1	0	0	-216.52	0	0
		2	0	0	0	0	0
		3	0	0	-639.65	0	0
		4	0	0	-1117.96	0	0
		5	0	0	-1503.6	0	0
		6	0	0	-51	0	0
		7	0	0	-81.1	0	0
		8	0	0	-118.4	0	0
		9	21.03	0	0	0	628.19
		10	-21.03	0	0	0	-628.19
		11	0	21.03	0	-628.19	0
		12	0	-21.03	0	628.19	0
		13	278.14	0	0	0	8673.58
		14	0	278.14	0	-8673.58	0
		15	172.87	0	0	0	5758.56
		16	0	172.87	0	-5758.56	0
	2	1	0	0	-3.84	0	0
		2	0.58	0.32	-108.59	-1.69	0
		3	0.06	0.12	-7.9	-0.59	0
		4	0.08	0.16	-13.2	-0.81	0
		5	0.06	0.16	-21.61	-0.83	0
		6	0	0	0	0	0
		7	0.08	0.16	-13.2	-0.81	0
		8	0.06	0.16	-21.61	-0.83	0
		9	-1.76	3.11	-4.3	12.91	0
		10	1.76	-3.11	4.3	-12.9	0
		11	19.27	1.11	-18.27	3.98	0
		12	-19.27	-1.11	18.27	-3.98	0
		13	-13.52	15.07	-21.64	66.7	0
		14	46.1	1.42	-47.35	4.25	0
		15	0	0	0	0	0



Calculation Sheet of 121-D Foundation

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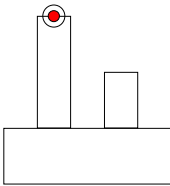
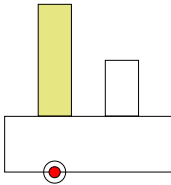
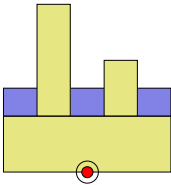
Project No. : 12-1812

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3	16	0	0	0	0	0
	1	0	0	-3.84	0	0
	2	-0.03	-0.01	-174.87	0	0
	3	0.02	-0.01	-23.77	0	0
	4	0.04	-0.02	-35.05	0	0
	5	0.04	-0.03	-43.21	0	0
	6	0	0	0	0	0
	7	0.61	0.93	-48.51	0	0
	8	0.67	1.03	-52.1	0	0
	9	-0.03	3.08	-2.49	0	0
	10	0.03	-3.08	2.49	0	0
	11	3.23	1.03	-19.77	0	0
	12	-3.23	-1.03	19.77	0	0
	13	-0.19	15.03	-21.15	0	0
	14	7.02	1.23	-51.29	0	0
	15	0	0	0	0	0
	16	0	0	0	0	0
Footing SW		0.000	0.000	-616.571	0.000	0.000

3.4 LOAD COMBINATION

In Pier Top without Self Weight 	In Footing Bottom with Pier Self Weight, But without Footing Self Weight, 	In Footing Bottom Center with Pier & Footing Self Weight & Soil Weight, Case PileType in centroid of Pile Group Case NonPileType in centroid of Footing 
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3.4.1 Load Combination in Pier Top (Without SW)

Unit(tonf , tonf-m)

Ft.Name	Pr.Name	L.Comb.	SFx	SFy	SFz	SMx	SMy
F6	1	101	0.000	0.000	-690.650	0.000	0.000
		102	12.618	0.000	-690.650	0.000	376.914
		103	-12.618	0.000	-690.650	0.000	-376.914
		104	0.000	12.618	-690.650	-376.914	0.000
		105	0.000	-12.618	-690.650	376.914	0.000
		106	121.009	36.303	-690.650	-1209.298	4030.992
		107	121.009	-36.303	-690.650	1209.298	4030.992
		108	-121.009	36.303	-690.650	-1209.298	-4030.992
		109	-121.009	-36.303	-690.650	1209.298	-4030.992
		110	36.303	121.009	-690.650	-4030.992	1209.298
		111	36.303	-121.009	-690.650	4030.992	1209.298
		112	-36.303	121.009	-690.650	-4030.992	-1209.298
		113	-36.303	-121.009	-690.650	4030.992	-1209.298
		114	12.618	0.000	-414.390	0.000	376.914
		115	-12.618	0.000	-414.390	0.000	-376.914
		116	0.000	12.618	-414.390	-376.914	0.000
		117	0.000	-12.618	-414.390	376.914	0.000
		118	121.009	36.303	-414.390	-1209.298	4030.992



Calculation Sheet of 121-D Foundation

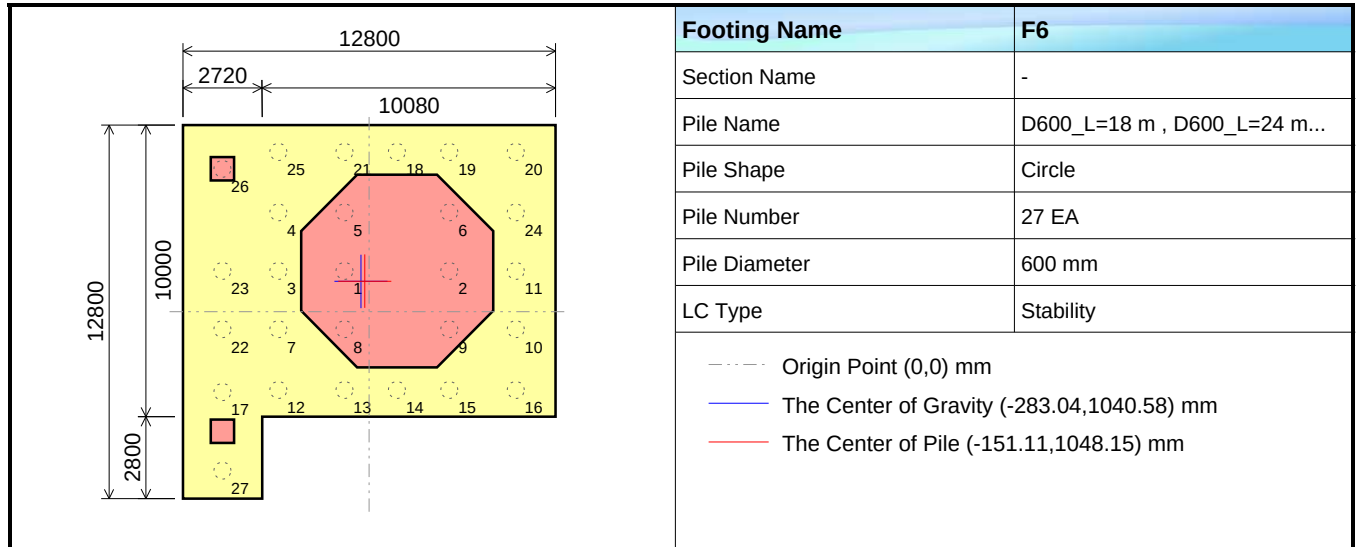
Project Na. : PUSRI IIB

Project No. : 12-1812

Client : PT PUSRI PLB

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3.4.4 Pile Reaction Table



• LC : 101, (1.0 SW + 1.0 DL + 1.0 E(e) : Equipment Load (Erection) + 1.0 P(e) : Piping Load (Erection))

Unit (mm,tonf)

No.	Name	Pile Geometry		Bi-Axial	Shear (Hor)	Ra	Ua	Ha
		X	Y	XY-Dir.	XY-Dir.			
1	D600_L=18 m	-840	1400	69.35	0.03	225	26	23.8
2	D600_L=18 m	2760	1400	62.59	0.03	225	26	23.8
3	D600_L=18 m	-3120	1400	73.63	0.03	225	26	23.8
4	D600_L=18 m	-3120	3400	71.74	0.03	225	26	23.8
5	D600_L=18 m	-840	3400	67.46	0.03	225	26	23.8
6	D600_L=18 m	2760	3400	60.7	0.03	225	26	23.8
7	D600_L=24 m	-3120	-600	75.52	0.03	135	54	23.8
8	D600_L=26/2	-840	-600	71.24	0.03	166.5	64.8	23.8
9	D600_L=30 M	2760	-600	64.48	0.03	225	91.33	23.8
10	D600_L=30m	5040	-600	60.2	0.03	225	96	23.8
11	D600_L=30m	5040	1400	58.31	0.03	225	96	23.8
12	D600_L=30m	-3120	-2680	77.49	0.03	225	96	23.8
13	D600_L=30m	-840	-2680	73.21	0.03	225	96	23.8
14	D600_L=30m	960	-2680	69.83	0.03	225	96	23.8
15	D600_L=30m	2760	-2680	66.45	0.03	225	96	23.8
16	D600_L=30m	5040	-2680	62.17	0.03	225	96	23.8
17	D600_L=30m	-5040	-2750	81.16	0.03	225	96	23.8
18	D600_L=30m	960	5480	62.11	0.03	225	96	23.8
19	D600_L=30m	2760	5480	58.73	0.03	225	96	23.8
20	D600_L=30m	5040	5480	54.45	0.03	225	96	23.8
21	D600_L=30m	-840	5480	65.5	0.03	225	96	23.8
22	D600_L=30m	-5040	-600	79.13	0.03	225	96	23.8
23	D600_L=30m	-5040	1400	77.24	0.03	225	96	23.8
24	D600_L=26/2	5040	3400	56.42	0.03	166.5	68.96	23.8



Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

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Client : PT PUSRI PLB

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4. CHECK OF STABILITY

4.1 CHECK OF PILE REACTION (Bi-Axial)

4.1.1 Formula

※ if footing is checked in Buoyancy ΣFz means $\Sigma Fz - Fb$

$$a. \text{ Vertical - Bi Axial : } R = \frac{\Sigma Fz}{N_p} \pm \frac{\Sigma My \times X}{\Sigma X_i^2} \pm \frac{\Sigma Mx \times Y}{\Sigma Y_i^2}$$

$$- R_u = R_{\max}$$

$$- U_f = \text{Min}[0, R_{\min}]$$

$$- R_u < V_a \rightarrow \text{OK}$$

$$b. \text{ Horizontal - } H_{\max} = \frac{\sqrt{(\Sigma Hx_i^2 + \Sigma Hy_i^2)}}{N_p} < H_a \rightarrow \text{OK}$$

$$c. \text{ Uplift - } U_f < U_a \rightarrow \text{OK}$$

$$\text{Ver. / } U_f = \text{Vertical / Uplift}$$

4.1.2 Check of Vertical & Uplift Reaction

Ft.Name	Np(EA)	FI (mm)	Fw (mm)	SXi ² (m ²)	SYi ² (m ²)
F6	27	12800	12800	345.53	275.52

Unit(tonf)

Ft.Name	L.Comb.	Pile	R _{Max}	R _{Min}	R _u	U _f	R _a	U _a	Result
F6	101	D600_L=18	73.634	60.7	73.634	0	225	26	OK
		D600_L=24	75.525	75.525	75.525	0	135	54	
		D600_L=26/	71.243	71.243	71.243	0	166.5	64.8	
		D600_L=30	64.481	64.481	64.481	0	225	91.33	
		D600_L=30	81.163	54.452	81.163	0	225	96	
		D600_L=26/	73.932	56.418	73.932	0	166.5	68.96	
		D600_L30m	83.716	83.716	83.716	0	225	96.28	
	102	D600_L=18	70.323	64.34	70.323	0	299.25	34.58	OK
		D600_L=24	72.143	72.143	72.143	0	179.55	71.82	
		D600_L=26/	70.529	70.529	70.529	0	221.445	86.184	
		D600_L=30	67.98	67.98	67.98	0	299.25	121.469	
		D600_L=30	75.459	60.833	75.459	0	299.25	127.68	
		D600_L=26/	68.498	62.726	68.498	0	221.445	91.717	
		D600_L30m	77.916	77.916	77.916	0	299.25	128.052	
	103	D600_L=18	76.945	57.059	76.945	0	299.25	34.58	OK
		D600_L=24	78.906	78.906	78.906	0	179.55	71.82	
		D600_L=26/	71.956	71.956	71.956	0	221.445	86.184	
		D600_L=30	60.982	60.982	60.982	0	299.25	121.469	
		D600_L=30	86.868	48.069	86.868	0	299.25	127.68	
		D600_L=26/	79.366	50.109	79.366	0	221.445	91.717	
		D600_L30m	89.515	89.515	89.515	0	299.25	128.052	
	104	D600_L=18	76.595	63.489	76.595	0	299.25	34.58	OK
		D600_L=24	74.384	74.384	74.384	0	179.55	71.82	
		D600_L=26/	69.731	69.731	69.731	0	221.445	86.184	
		D600_L=30	62.384	62.384	62.384	0	299.25	121.469	
		D600_L=30	79.408	56.582	79.408	0	299.25	127.68	
		D600_L=26/	81.342	59.941	81.342	0	221.445	91.717	



Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

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Client : PT PUSRI PLB

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4.1.3 Check Of Horizontal Reaction

Ft.Name	L.Comb.	Pile	Hmax (tonf)	Ha (tonf)	Result
F6	101	D600_L=18 m	0.028	23.8	OK
		D600_L=24 m		23.8	
		D600_L=26/ 27m		23.8	
		D600_L=30 M		23.8	
		D600_L=30 m_per		23.8	
		D600_L=26/ 27m_per		23.8	
		D600_L30m _per_tensio		23.8	
	102	D600_L=18 m	0.476	31.654	OK
		D600_L=24 m		31.654	
		D600_L=26/ 27m		31.654	
		D600_L=30 M		31.654	
		D600_L=30 m_per		31.654	
		D600_L=26/ 27m_per		31.654	
		D600_L30m _per_tensio		31.654	
	103	D600_L=18 m	0.422	31.654	OK
		D600_L=24 m		31.654	
		D600_L=26/ 27m		31.654	
		D600_L=30 M		31.654	
		D600_L=30 m_per		31.654	
		D600_L=26/ 27m_per		31.654	
		D600_L30m _per_tensio		31.654	
	104	D600_L=18 m	0.745	31.654	OK
		D600_L=24 m		31.654	
		D600_L=26/ 27m		31.654	
		D600_L=30 M		31.654	



Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

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5. DESIGN OF FOOTING

5.1 DESIGN MOMENT AND SHEAR FORCE

Footing design is in accordance with unltimate strength method at footing bottom.

Calculated total pier load as

$$\Sigma Q = \Sigma F_z - \text{Self Weight Factor} \times (\text{Soil Weight} + \text{Footing Weight})$$

Ft.Name : Footing Name, Sec.Name : Strip Name for Footing Reinforcement Design

Dir. : Direction, L.Comb. : Load Combination Index, Fl or Fw : Footing Length or Footing Width, Sl or Sw : Strip X or Y width

5.1.1 Data

Unit(mm , tonf , tonf-m)

Ft.Name	Sec.Nam	Dir.	L.Comb.	Fl or Fw	Sl or Sw	SFz	SM	SQ
F6 ①	S1	X	201	12800.00	12800.00	3116.787	-978.70	1721.972
			202	12800.00	12800.00	2678.320	-165.00	1482.765
			203	12800.00	12800.00	2664.745	-1512.77	1469.189
			204	12800.00	12800.00	2709.569	-932.55	1514.014
			205	12800.00	12800.00	2633.499	-745.23	1437.943
			206	12800.00	12800.00	2743.911	8630.33	1548.355
			207	12800.00	12800.00	2671.531	5628.44	1475.976
			208	12800.00	12800.00	2671.531	-7306.21	1475.976
			209	12800.00	12800.00	2671.531	-7306.21	1475.976
			210	12800.00	12800.00	2671.531	1101.32	1475.976
			211	12800.00	12800.00	2671.531	1101.32	1475.976
			212	12800.00	12800.00	2671.531	-2779.08	1475.976
			213	12800.00	12800.00	2671.531	-2779.08	1475.976
			226	12800.00	12800.00	2010.438	44.72	1113.771
			227	12800.00	12800.00	1996.862	-1303.05	1100.195
			228	12800.00	12800.00	2041.687	-722.83	1145.020
			229	12800.00	12800.00	1965.615	-535.51	1068.949
			230	12800.00	12800.00	2003.648	5838.17	1106.982
			231	12800.00	12800.00	2003.648	5838.17	1106.982
			232	12800.00	12800.00	2003.648	-7096.49	1106.982
			233	12800.00	12800.00	2003.648	-7096.49	1106.982
			234	12800.00	12800.00	2003.648	1311.04	1106.982
			235	12800.00	12800.00	2003.648	1311.04	1106.982
			236	12800.00	12800.00	2003.648	-2569.36	1106.982
			237	12800.00	12800.00	2003.648	-2569.36	1106.982
			238	12800.00	12800.00	3375.568	-616.81	2180.012
			239	12800.00	12800.00	3382.357	57.07	2186.801
			240	12800.00	12800.00	3368.781	-1290.70	2173.225
			241	12800.00	12800.00	3413.605	-710.48	2218.050
			242	12800.00	12800.00	3337.535	-523.16	2141.979
			243	12800.00	12800.00	3447.947	8852.41	2252.391
			244	12800.00	12800.00	3388.760	9011.09	2193.204
			245	12800.00	12800.00	3362.375	-10244.71	2166.819
			246	12800.00	12800.00	3303.188	-10086.02	2107.633
			247	12800.00	12800.00	3487.048	1983.28	2291.493



Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

Project No. : 12-1812

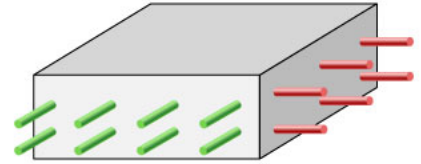
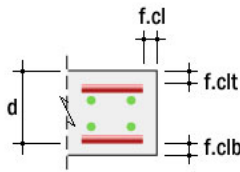
Client : PT PUSRI PLB

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5.1.2 Design Parameters

Yield Strength - D10 ~ D16 : f_y1 , D19 ~ : f_y2 f_{cl} : Clear Cover for edge of footing reinforcement f_{clt} : Clear Cover for top of footing reinforcement f_{p_clb} : Clear Cover for bottom of footing reinforcement

Loc. : Location of Critical Point from left side of footing

Unit(kgf/cm²,mm)

f(Flexure)	f(Shear)	f'c	fy1	fy2	f_cl	f_clt	f _{p_clb}
0.85	0.8	210.00	40000.00	40000.00	75.0	75.0	75.0



Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

Project No. : 12-1812

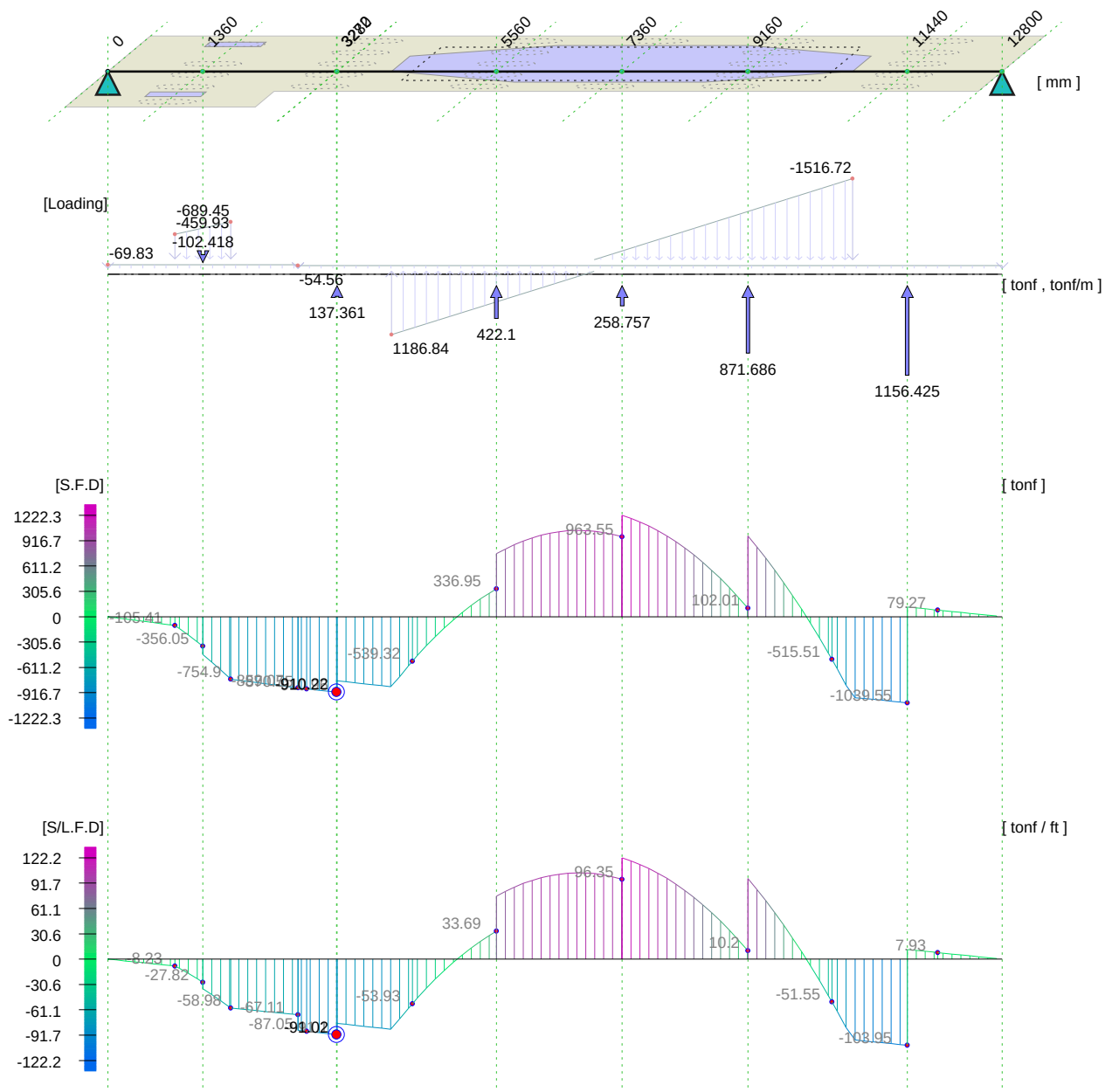
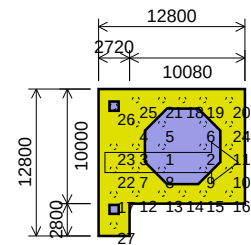
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Title

Shear Force Diagram

Foundation name	F6	Section name	S1	Direction	X	L/C ID	206
Analysis Method Conventional Rigid Method with equilibrium equation (Method 2)							
ΣF_z	-2743.911 tonf	ΣM_y	8630.334 tonf-m	Moment inertia		1934.9170 m ⁴	
Area	135.616 m ²	Contact Area		Critical Point Method		Critical Point	
Critical Value Vu = -91.022 tonf/ft							





Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

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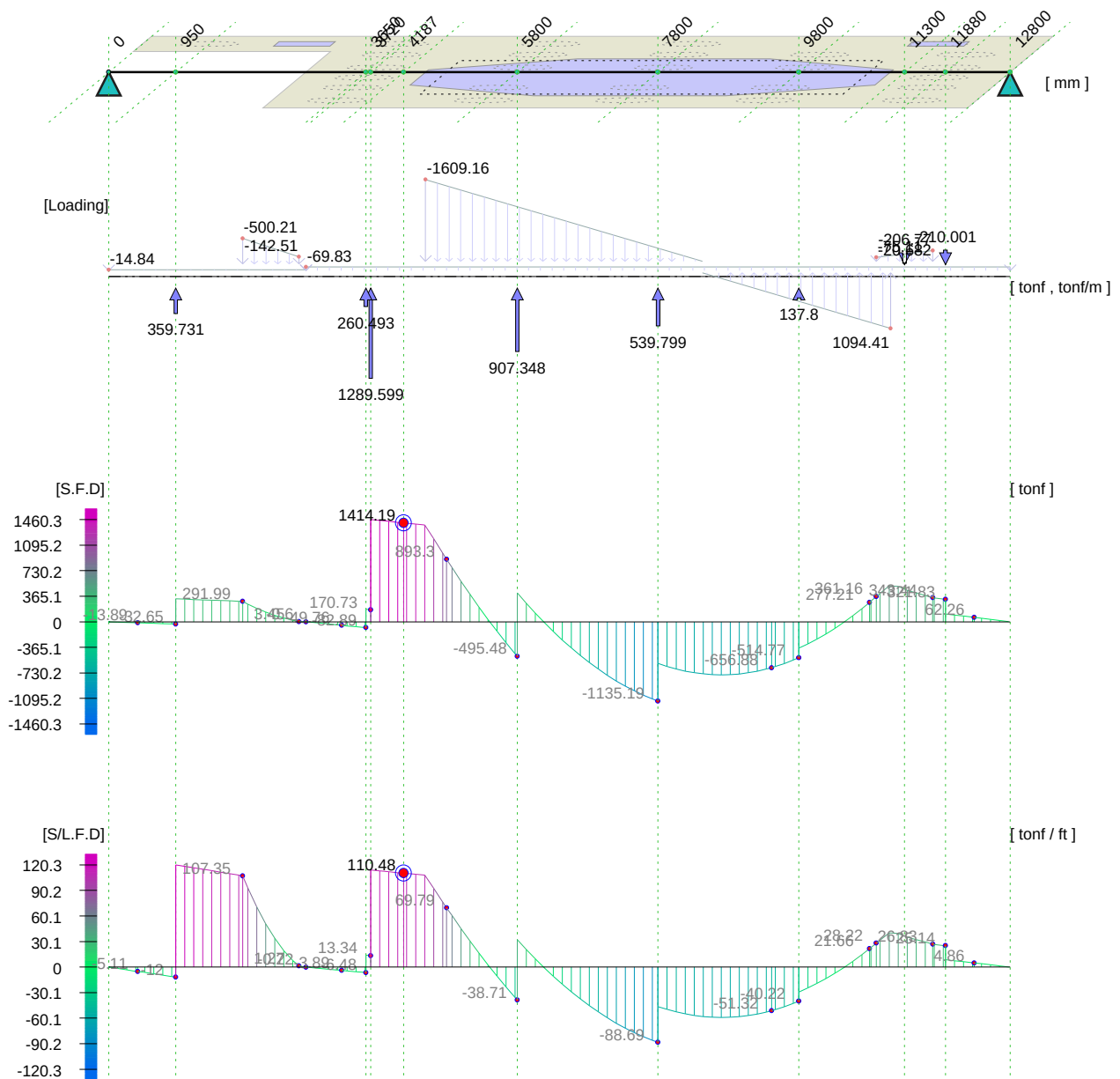
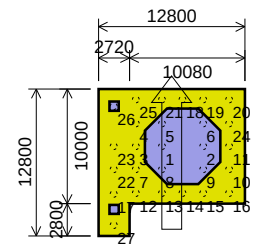
Client : PT PUSRI PLB

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Title

Shear Force Diagram

Foundation name	F6	Section name	S1	Direction	Y	L/C ID	250
Analysis Method Conventional Rigid Method with equilibrium equation (Method 2)							
ΣF_z	-3264.087 tonf	ΣM_x	10126.580 tonf-m	Moment inertia		1366.0750 m ⁴	
Area	135.616 m ²	Contact Area		Critical Point Method		Critical Point	
Critical Value Vu = 110.484 tonf/ft							





Calculation Sheet of 121-D Foundation

Project Na. : PUSRI IIB

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5.2 PILE PUNCHING SHEAR FORCE

5.2.1 Pile Punching Shear Formula

$$V_u = \Sigma F_z \cdot \text{Shade Ratio}$$

$$(a) \phi V_{c1} = 0.8 \cdot 2 \cdot (1 + 2/\beta_c) \sqrt{f_{ck}} b_o \cdot d \quad (\text{eq 11-33}) \quad <- V_{c1}$$

$$(b) \phi V_{c2} = 0.8 \cdot 2 \cdot (1 + \alpha_s d / 2 b_o) \sqrt{f_{ck}} b_o \cdot d \quad (\text{eq 11-34}) \quad <- V_{c2}$$

$$(c) \phi V_{c3} = 0.8 \cdot 4 \sqrt{f_{ck}} b_o \cdot d \quad (\text{eq 11-35}) \quad <- V_{c3}$$

$$\phi V_c = \text{Min}(\phi V_{c1}, \phi V_{c2}, \phi V_{c3}) \quad \text{ACI 318-05 CODE 11.12.2.1}$$

$V_u \leq \phi V_c$, then OK

where

β = ratio of long side to short side of the column, concentrated load or reaction area

α_s = 40 for interior columns

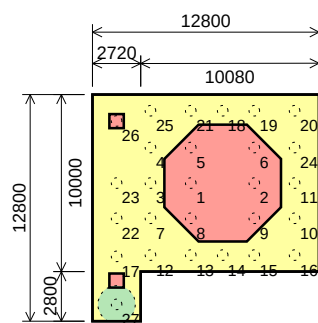
= 30 for edge columns

= 20 for corner columns

b_o = perimeter of critical section

$$\text{Shade Ratio} = \frac{\text{Footing Area} - \text{Punching Area}}{\text{Footing Area}}$$

5.2.2 Check of Pile Punching Shear



Ft.Name	F6	Punching Area	34189.060 cm ²
Pile No.	27	bc / as	1 / 30
Shape	Circle	f Vc1	1586.473 tonf
L.Comb.	250	f Vc2	2637.885 tonf
PileName	D600_L30m_per_tension	f Vc3	1057.648 tonf
Diameter	600mm	f Vc	1057.648 tonf
bo	5687.89mm	Vu	387.438 tonf
d	1512.3mm	Result	OK